

Mathematical Models & Formulation Reference

Urban Population Forecasting & Resource Allocation System | Greater Chennai Corporation (15 Zones)

Document Overview: This reference compiles the complete mathematical, machine learning, and municipal engineering formulations implemented across the data pipeline, time-series neural networks, spatial localizer, civic infrastructure demand models, and multi-criteria vulnerability priority scoring framework.

1. Demographic & Earth Observation Feature Engineering

1.1 Temporal Linear Interpolation (Inter-Census & GHSL Epoch Alignment)

$$y(t) = y(t_1) + \left[\frac{(t - t_1)}{(t_2 - t_1)} \right] * [y(t_2) - y(t_1)]$$

Description: Linearly bridges observational gaps between Global Human Settlement Layer (GHSL) satellite survey epochs.

Parameters: t: Target estimation year | t1, t2: Anchor sensor survey years | y(t): Estimated built-up surface area

1.2 Built-Up Surface Area Metric Conversion

$$\text{Built-Up Area (sq.km)} = \text{Built-Up Surface Area (sq.m)} / 1,000,000$$

Description: Converts Copernicus GHSL raster pixel square-meter counts to standard municipal planning square-kilometer units.

1.3 Annual Demographic Growth Rate

$$\text{growth_rate}(t) = \left[\frac{\text{Population}(t) - \text{Population}(t-1)}{\text{Population}(t-1)} \right] * 100\%$$

Description: Percentage demographic expansion calculated year-over-year.

Parameters: Population(t): Zone population at year t | Population(t-1): Population at previous year

1.4 Dynamic Satellite Proxies (Built-Up & Night-Light Deltas)

$$\begin{aligned} \text{Delta_BuiltUp}(t) &= \text{BuiltUp}(t) - \text{BuiltUp}(t-1) \text{ [sq.m / year]} \\ \text{Delta_NightLight}(t) &= \text{NightLight}(t) - \text{NightLight}(t-1) \text{ [nW / (cm}^2 * \text{sr)]} \end{aligned}$$

Description: Physical expansion differentials tracking structural urbanization and night-time radiometric radiance shifts.

2. Machine Learning & Forecasting Model Architectures

2.1 Baseline Linear Regressor (Ordinary Least Squares)

$$\begin{aligned} P_{\text{pred}}(t) &= \text{beta}_0 + \text{beta}_1 * t \\ \text{where beta} &= (X^T * X)^{-1} * X^T * y \end{aligned}$$

Description: Serves as benchmark linear baseline extrapolating historical trends.

2.2 Deep Long Short-Term Memory (LSTM) Cell Recurrence

```
f_t = sigmoid( W_f * [h_{t-1}, x_t] + b_f ) [Forget Gate]
i_t = sigmoid( W_i * [h_{t-1}, x_t] + b_i ) [Input Gate]
C_cand_t = tanh( W_c * [h_{t-1}, x_t] + b_c ) [Candidate State]
C_t = f_t * C_{t-1} + i_t * C_cand_t [Cell Memory Update]
o_t = sigmoid( W_o * [h_{t-1}, x_t] + b_o ) [Output Gate]
h_t = o_t * tanh( C_t ) [Hidden State Vector]
```

Description: PyTorch sequential neural network learning multi-year demographic momentum and memory vectors.

Parameters: sigmoid(z) = $1 / (1 + \exp(-z))$ | tanh: Hyperbolic tangent | h_t : Hidden state output vector

2.3 Hybrid Machine Learning Pipeline (LSTM + XGBoost Spatial Calibration)

```
P_Hybrid(t) = P_LSTM(t) + SUM_{m=1..M} f_m( z_t )
z_t = [ P_LSTM(t), BuiltUp(t), NightLight(t), Delta_BuiltUp, Delta_NightLight,
GrowthRate, Zone_ID ]
Objective: L = SUM ( y_i - y_pred_i )^2 + SUM [ gamma * T_m + 0.5 * lambda *
||w_m||^2 ]
```

Description: Combines temporal sequence prediction with gradient boosted regression trees to account for spatial satellite proxies.

Parameters: M: Number of boosted trees | T_m : Leaf count | gamma, lambda: Complexity and L2 penalties

3. Statistical Model Performance Evaluation Metrics

Metric	Mathematical Formulation	Benchmark Value (Hybrid)
Mean Absolute Error (MAE)	$MAE = (1/N) * \sum y_i - y_{pred_i} $	4,337.70 persons (47.7% reduction vs LSTM)
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{(1/N) * \sum (y_i - y_{pred_i})^2}$	6,440.19 persons
Mean Absolute % Error (MAPE)	$MAPE = (100\% / N) * \sum (y_i - y_{pred_i}) / y_i $	0.66%
Coefficient of Determination (R^2)	$R^2 = 1 - [\sum (y_i - y_{pred_i})^2 / \sum (y_i - y_{mean})^2]$	0.9994 (99.94% variance explained)

4. Municipal Civic Resource Demand Forecasting

4.1 Municipal Potable Water Demand (CPHEEO Guideline)

```
Water Demand (LPD) = Population_z * 135.0 [Liters / Person / Day]
Water Demand (MLD) = [ Population_z * 135.0 ] / 1,000,000 [Million Liters / Day]
```

Description: Derived from Indian Central Public Health and Environmental Engineering Organisation (CPHEEO) standard for metropolitan cities with piped sewerage.

Parameters: Population_z: Projected Population in Zone z | 1 MLD = 1,000,000 Liters/day

4.2 Electric Power Grid Consumption Demand (CEA Standard)

$$\text{Power Demand (kWh/day)} = \text{Population}_z * 3.5 \text{ [kWh / Person / Day]}$$
$$\text{Grid Energy Demand (MWh/day)} = [\text{Population}_z * 3.5] / 1,000 \text{ [MegaWatt-Hours / Day]}$$

Description: Based on Central Electricity Authority (CEA) per-capita urban domestic and commercial consumption parameters.

Parameters: 1 MWh = 1,000 kWh

4.3 Civic Social Infrastructure: Healthcare Capacity (WHO Standard)

$$\text{Hospital Beds Required} = \text{Population}_z * 0.003 = [\text{Population}_z * 3] / 1,000$$

Description: World Health Organization (WHO) municipal benchmark of 3 acute hospital beds per 1,000 residents.

4.4 Civic Social Infrastructure: Education Capacity (UDPFI Guideline)

$$\text{School Seats Required} = \text{Population}_z * 0.050 = [\text{Population}_z * 50] / 1,000$$

Description: Urban Development Plans Formulation and Implementation (UDPFI) benchmark of 50 school seats per 1,000 residents for basic cohort schooling.

5. Multi-Criteria Priority Vulnerability Score Decomposition

5.1 Component Factor Definitions

1. **Demographic Growth Factor:** $G_z = \max(0.0, \text{growth_rate}_{\%} * 2.5)$
 2. **Built-Up Density Factor:** $D_z = (\text{Population}_z / 100,000) * 5.0$
 3. **Resource Demand Factor:** $R_z = (\text{WaterDemand_LPD} / 10,000,000) * 3.0$

Description: Dimensionless indicators capturing demographic surge, physical spatial crowding, and utility load deficits.

5.2 Composite Priority Vulnerability Index (MCVI)

$$\text{Priority Score (V}_z\text{)} = (0.40 * G_z) + (0.35 * D_z) + (0.25 * R_z)$$
$$V_z = \text{Growth_Contribution} + \text{Density_Contribution} + \text{Demand_Contribution}$$

Description: Weighted multi-criteria decision analysis prioritizing municipal budget allocation.

Parameters: w_g = 0.40 (Growth) | w_d = 0.35 (Built-Up Density) | w_r = 0.25 (Utility Demand)

Score Range (V_z)	Priority Tier	Municipal Policy Directive
V_z >= 35.0	CRITICAL	Immediate multi-agency emergency infrastructure capital allocation mandated before 2030 horizon.
25.0 <= V_z < 35.0	HIGH	Top priority for trunk water main & substation capacity augmentation in next budgetary cycle.
15.0 <= V_z < 25.0	MEDIUM	Monitored capacity growth; routine phased civic infrastructure upgrades.
V_z < 15.0	LOW	Adequately provisioned; standard municipal maintenance schedule.